There Is No 11-Arc in the Projective Plane of Order 8

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Standard projective-plane interpretation

An arc in $\mathrm{PG}(2, q)$ is a set of points with no three collinear. In a projective plane of order q, exactly $q+1$ lines pass through each point. If an arc has k points, fix one point P. Each of the other $k-1$ points lies on a line through P, and these lines must be distinct: two other points on the same line would make three arc points collinear. Therefore $k-1 \leq q+1$, or $k \leq q+2$.

For $q = 8$, this gives $k \leq 10$. There are no arcs of length 11 in $\mathrm{PG}(2, 8)$, so the number of their isomorphism classes is zero, not one. The conjecture does not name an ambient projective space; this disproof uses the standard projective-plane meaning of "MDS arc," consistent with the surrounding arc conjectures in the repository.

Formalization scope

Lean checks the decisive cardinality contradiction: an 11-arc would require ten distinct lines through a fixed point, while an order-8 plane has only nine. The projective-plane incidence facts and the no-three-collinear injection are proved in the text and are not formalized in Lean.

Lean formalization

Lean formalizes the final finite counting contradiction (10 distinct lines would have to fit among 9 lines). The projective-plane incidence argument supplying those counts is explicit in the report.